課前準備

- 本次課程會用到三個AI影像分割軟體: ilastik, Cellpose-SAM, micro-SAM, 安裝步驟請參見後方投影片

- 請各位學員嘗試安裝, 其中Cellpose 和micro-SAM需要相當長的安裝時間, 請耐心等待!

  (Tips: 如果你和我一樣沒耐心, 在miniforge prompt 裡面的執行conda 都可以改用mamba, 會快很多! 如果使用Anaconda的話, 只能用conda)

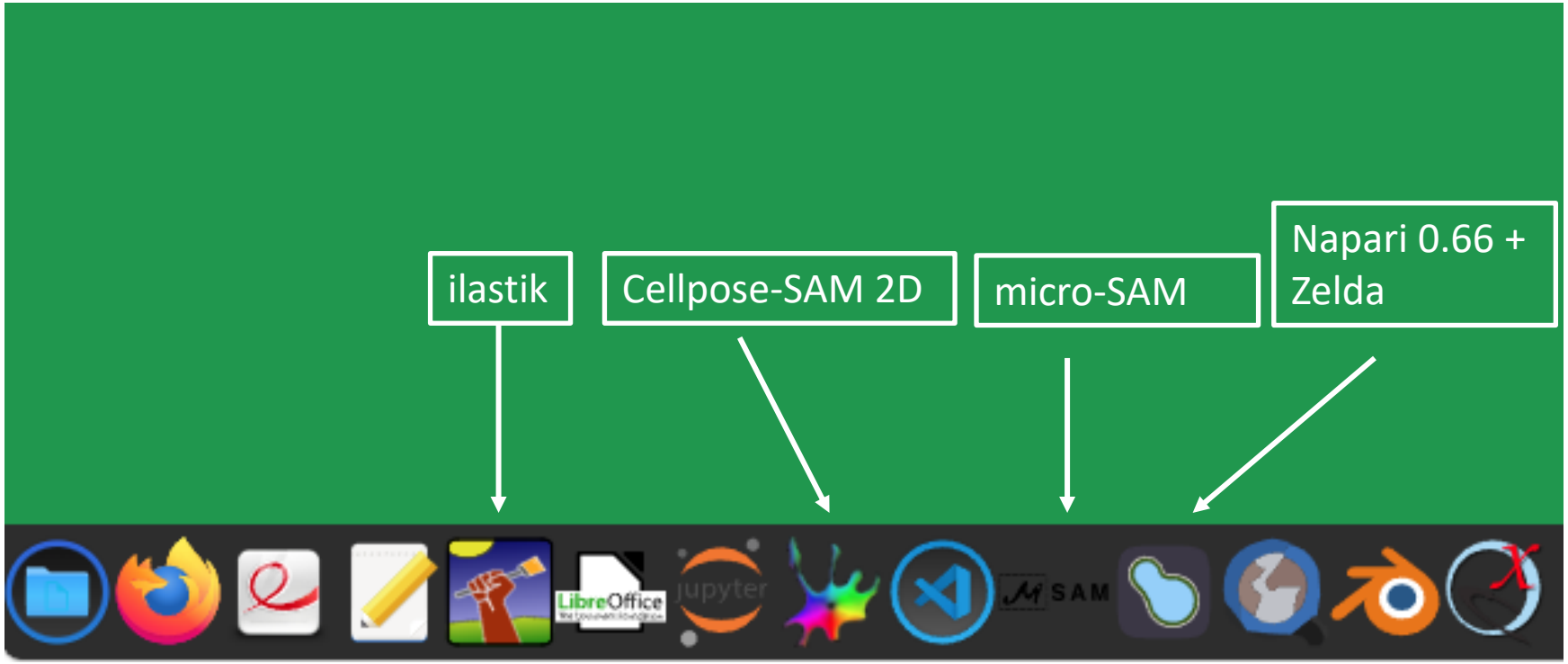
- Alternative: 在BARD平台上面有預先裝好這些軟體，請使用上課發的帳號登入。 https://bard-external.embl.de/

You can use all the tools on BARD or in a local installation

BARD

EMBL NFDI 4 BIOIMAGE deNBI

- This project is partly funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under the National Research Data Infrastructure – NFDI 46/1 – 501864659
- This project is also supported by EMBL and deNBI
- We would like to acknowledge the previouse work done by the Orange S.A.



Supported by Yi Sun @EMBL

https://bard-external.embl.de/

ilastik installation

請下載並安裝 **1.4.1 Post1**穩定版, 如有**Nvidia GPU** 可選**GPU**版本

(Stable version)

Release 1.4.1.post1 (31. May 2025)

Release Highlights:

- Introduces support for the OME-Zarr file format
- Added Trainable Domain Adaptation Workflow
- Native build for Apple Silicon (M1, M2, M3, M4) and Intel Macs
- Tiff reading improvements
- Added spherical texture object features, contributed by Aafke Gros (publication)
- Better user interface in dark mode

Notes:

- Native builds for Apple Silicon (M1, M2, M3, M4) and Intel Macs are available, which means better performance on these machines overall. Notes for the silicon version:
  - Currently only 2D networks are run accelerated on apple hardware, leveraging Metal Performance Shaders (mps), in the Neural Network and Trainable Domain Adaptation Workflows. 3D networks will fallback to CPU execution.
- Commercial solver support with gurobi `12.0.0` for tracking with learning.

CPU

Regular builds

Windows (64-bit)	version 1.4.1.post1 (534 MB)	**Supported Versions:** Windows 10 and up
Apple Silicon (arm64), OSX	version 1.4.1.post1 (520 MB)	**Supported Versions:** 11.0 and up
Apple intel (64-bit), OSX	version 1.4.1.post1 (776 MB)	**Supported Versions:** 10.9 and up
Linux (64-bit)	version 1.4.1.post1 (786 MB)	**Supported Versions:** Ubuntu 14 and up

GPU

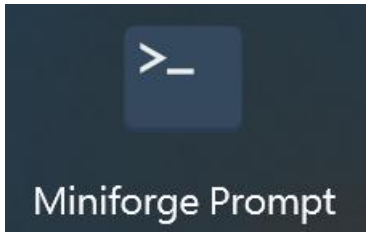
GPU-enabled builds

For faster neural network prediction. ilastik GPU-enabled builds are distributed with NVidia's CUDA toolkit. By downloading and using these builds, you accept the terms and conditions of the CUDA End User License Agreement (EULA).

Windows (64-bit)	version 1.4.1.post1-gpu (1.9 GB)	**Supported Versions:** Windows 10 and up, comes with cudatoolkit 11.8
Linux (64-bit)	version 1.4.1.post1-gpu (3.4 GB)	**Supported Versions:** Ubuntu 14 and up, comes with cudatoolkit 11.8

下載網址: https://www.ilastik.org/download

Micro-SAM installation



1. 利用Miniforge 建立給micro-sam 專用的Python 環境

```
conda create –n micro_sam-env
```

2. 啟動該micro-sam的環境

```
conda activate micro_sam-env
```

3. 安裝micro-sam

```
conda install –c conda-forge micro_sam
```

4. 啟動napari, 在plugin選單中可以找到Segment Anything for Microscopy

https://computational-cell-analytics.github.io/micro-sam/micro_sam.html#from-conda

Cellpose-SAM 安裝步驟

1. 利用Miniforge 建立給Cellpose 專用的Python 環境
   conda create --name cellpose python=3.10
2. 啟動該Cellpose的環境
   conda activate cellpose

3. 安裝Cellpose
   python -m pip install cellpose[gui]

CPU 版

GPU 版通常需要額外步驟

(4) 反安裝 PyTorch
pip uninstall torch

(5) 安裝 GPU版 PyTorch (建議安裝之前先更新Nvidia GPU Driver和CUDA toolkit)
pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cu126

6. 開啟cellpose GUI
   cellpose (2D) or cellpose --Zstastk (3D)

也可參考舊版的安裝影片, 步驟類似:

Cellpose 安裝全攻略 | AI 細胞分割工具安裝與設定教學 - YouTube

Cellpose installation (CPU-version)

Cellpose 安裝步驟原文說明

Option 1: Installation Instructions with conda

If you have an older `cellpose` environment you can remove it with `conda env remove -n cellpose` before creating a new one.

If you are using a GPU, make sure its drivers and the cuda libraries are correctly installed.

1. Install a miniforge distribution of Python. Note you might need to use an anaconda prompt if you did not add anaconda to the path.
2. Open an anaconda prompt / command prompt which has `conda` for **python 3** in the path
3. Create a new environment with `conda create --name cellpose python=3.10`. We recommend python 3.10, but python 3.9 and 3.11 will also work.
4. To activate this new environment, run `conda activate cellpose`
5. (option 1) To install cellpose with the GUI, run `python -m pip install cellpose[gui]`. If you're on a zsh server, you may need to use ' ': `python -m pip install 'cellpose[gui]'`.
6. (option 2) To install cellpose without the GUI, run `python -m pip install cellpose`.

如需不同版本, 請分開建立不同的Python 環境

Current Version: 4..1.1 (Cellpose-SAM)

If you want to specify the older version:
pip install "cellpose[gui]==3.1.1.2"

https://github.com/MouseLand/cellpose

Cellpose installation (GPU-version), optional

Cellpose 安裝步驟原文說明

GPU version (CUDA) on Windows or Linux

If you plan on running many images, you may want to install a GPU version of *torch*. To use your NVIDIA GPU with python, you will need to make sure the NVIDIA driver for your GPU is installed, check out this website to download it. You can also install the CUDA toolkit, or use the pytorch cudatoolkit (installed below with conda). If you have trouble with the below install, we recommend installing the CUDA toolkit yourself, choosing one of the 11.x releases here.

With the latest versions of pytorch on Linux, as long as the NVIDIA drivers are installed, the GPU version is installed by default with pip. You can check if the GPU support is working by opening the GUI. If the GPU is working then the `GPU` box will be checked and the `CUDA` version will be displayed in the command line.

If it's not working, we will need to remove the CPU version of torch:

```
pip uninstall torch
```

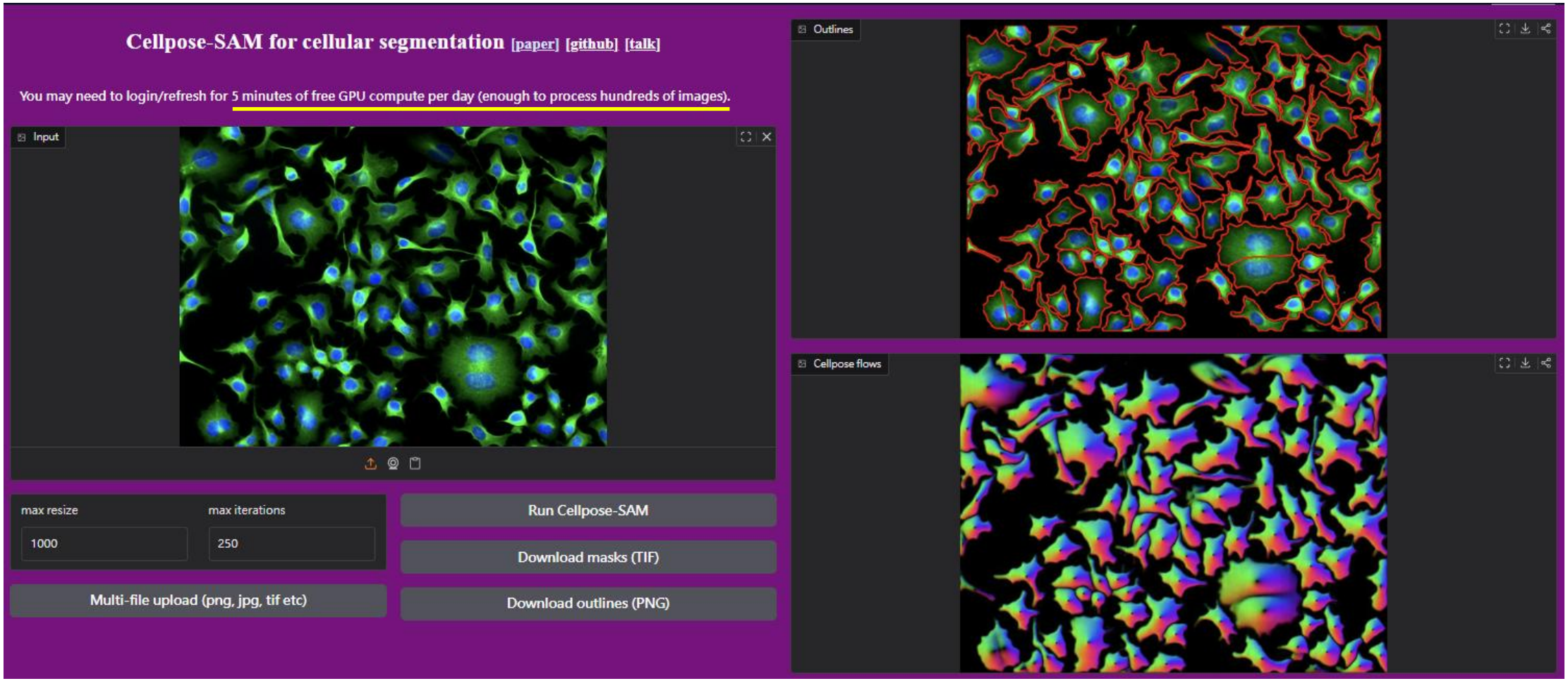
To install the GPU version of torch, follow the instructions here. The pip or conda installs should work across platforms, you will need torch and torchvision, e.g. for windows + cuda 12.6 the command is

```
pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cu126
```

Info on how to install several older versions is available here. After install you can check `conda list` for `pytorch`, and its version info should have `cuXX.X`, not `cpu`.

https://github.com/MouseLand/cellpose

Cellpose-SAM online (無須安裝, 可直接用, 但每日有上限)



也推薦看一下這個影片
用線上版玩看看

線上版 cellpose-sam與Fiji/imagej的結合應用 – YouTube (阿簡, 3 min video)

https://huggingface.co/spaces/mouseland/cellpose